CLASS WORK ONE (GROUPS)

INTRODUCTION TO BUSINESS ANALYTICS

OMIS 301

Topic: Understanding Data Types and Business Analytics through Health Data

This class work is to integrate theoretical and practical knowledge in business analytics, with a focus on applying analytical concepts to real-world health data. This assignment is divided into two parts: a theory component and a practical component.

Part 1: Theory (30 Marks)

Answer the following questions using the provided dataset. Provide brief but insightful responses based on the data.

Questions

1. **Data Types and Categories (5 Marks)**
   - Identify the types of data (nominal, ordinal, interval, ratio) for the following columns:
     - Country
     - Prevalence Rate (%)
     - Age Group
     - Per Capita Income (USD)
     - Year
   - Explain your reasoning for each data type classification.
2. **Supervised vs. Unsupervised Learning (5 Marks)**
   - If you were to build a predictive model for Recovery Rate (%), which columns in the dataset would serve as:
     - Predictors?
     - Target variable?
   - Would this task involve supervised or unsupervised learning? Justify your answer.
3. **Descriptive and Diagnostic Analytics (5 Marks)**

- Based on the dataset, what descriptive insights can you derive about the availability of healthcare across the countries?
- What diagnostic questions could you ask to explore disparities in Healthcare Access (%)?

4. **Data Mining and Patterns (5 Marks)**
    - Suggest a data mining approach to uncover patterns in the relationship between Urbanization Rate (%) and Disease Prevalence Rate (%).
    - What kind of insights could this approach potentially reveal?

5. **Correlation and Socio-Economic Factors (10 Marks)**
    - Explain how you would use correlation analysis to study the relationship between Education Index, Per Capita Income (USD), and Recovery Rate (%).
    - Based on the sample data, hypothesize potential relationships among these variables.

Part 2: Practical (70 Marks)

Instructions: Use Python for analysis. Submit your code, visualizations, and interpretations in a Jupyter Notebook.

Tasks

1. **Exploratory Data Analysis (20 Marks)**
    - Generate summary statistics for numerical and categorical columns in the dataset.
    - Visualize the distribution of Prevalence Rate (%) and Mortality Rate (%).
    - Create a bar chart comparing Healthcare Access (%) across the countries.
2. **Insights on Disease Prevalence and Healthcare Investment (20 Marks)**
    - Identify the top three diseases with the highest Prevalence Rate (%).
    - Suggest which countries require more investment in healthcare infrastructure based on Doctors per 1000 and Hospital Beds per 1000.
3. **Correlation Analysis (15 Marks)**
    - Calculate the correlation between Education Index, Per Capita Income (USD), and Prevalence Rate (%).
    - Visualize the relationships using scatter plots.

- Interpret the strength and direction of these relationships.

4. **Targeted Interventions and Public Health Campaigns (15 Marks)**
    - Identify regions (countries) with the highest Disease Burden (DALYs).
    - Recommend specific public health campaigns or interventions based on the data.
    - Use visualizations to support your recommendations.

Submission Guidelines

1. **Theory Part**: Submit a typed document (PDF format) with responses to the theory questions.
2. **Practical Part**: Submit a Jupyter Notebook with code, visualizations, and interpretations. Ensure your code is well-documented.

Deadline: Today 05 – 12 – 2024, 6 pm

Grading Criteria: Accuracy of answers, clarity of explanations, quality of visualizations, and depth of interpretations.

Dataset Information

This dataset provides comprehensive statistics on global health, focusing on various diseases, treatments, and outcomes. The data spans multiple countries and years, offering valuable insights for health research, epidemiology studies, and machine learning applications. The dataset includes information on the prevalence, incidence, and mortality rates of major diseases, as well as the effectiveness of treatments and healthcare infrastructure.

Column Descriptions

- **Country**: The name of the country where the health data was recorded.
- **Year**: The year in which the data was collected.
- **Disease Name**: The name of the disease or health condition tracked.
- **Disease Category**: The category of the disease (e.g., Infectious, Non-Communicable).
- **Prevalence Rate (%)**: The percentage of the population affected by the disease.
- **Incidence Rate (%)**: The percentage of new or newly diagnosed cases.
- **Mortality Rate (%)**: The percentage of the affected population that dies from the disease.

- **Age Group**: The age range most affected by the disease.
- **Gender**: The gender(s) affected by the disease (Male, Female, Both).
- **Population Affected**: The total number of individuals affected by the disease.
- **Healthcare Access (%)**: The percentage of the population with access to healthcare.
- **Doctors per 1000**: The number of doctors per 1000 people.
- **Hospital Beds per 1000**: The number of hospital beds available per 1000 people.
- **Treatment Type**: The primary treatment method for the disease (e.g., Medication, Surgery).
- **Average Treatment Cost (USD)**: The average cost of treating the disease in USD.
- **Availability of Vaccines/Treatment**: Whether vaccines or treatments are available.
- **Recovery Rate (%)**: The percentage of people who recover from the disease.
- **DALYs**: Disability-Adjusted Life Years, a measure of disease burden.
- **Improvement in 5 Years (%)**: The improvement in disease outcomes over the last five years.
- **Per Capita Income (USD)**: The average income per person in the country.
- **Education Index**: The average level of education in the country.
- **Urbanization Rate (%)**: The percentage of the population living in urban areas.

Best wishes!!